

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 21-12-25
Max. Marks: 300

21-12-25_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-4_SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

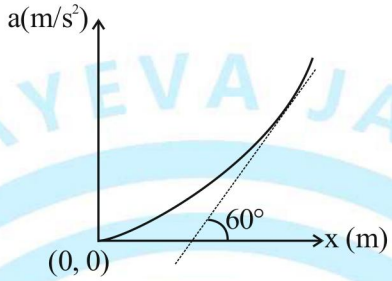
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

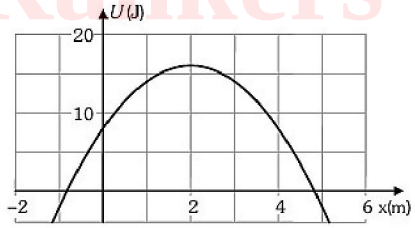
1. A particle starts moving with initial velocity 3 m/s along positive x-axis from the origin. Its acceleration is varying with position x in parabolic nature as shown in the figure. The slope of graph is zero at the origin. At $x = \sqrt{3}\,m$, tangent to the graph makes an angle 60° with positive x-axis as shown in diagram.

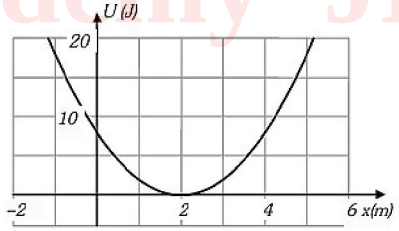


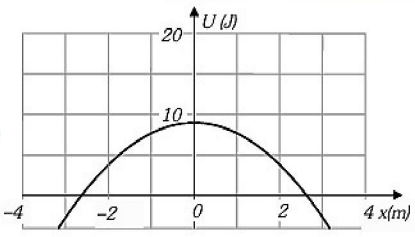
The acceleration of the particle at $x = \sqrt{3}\,m$ is

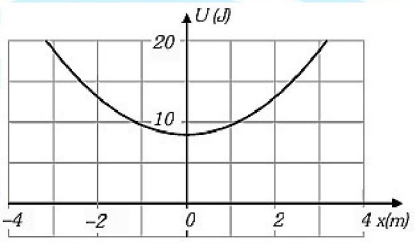
A) 1.0 m/s² B) 1.5 m/s² C) 2.5 m/s² D) 3.0 m/s²

2. A conservative force $\vec{F} = (4x - 8)\hat{i}$ N acts on a particle that moves on the x-axis. The potential energy associated with this force is assigned a value of 8 J at x = 0. Which of the following best represents variation in potential energy with position coordinate x?

A)


B)


C)


D)


3. A uniform chain of mass m forming a circle of radius R is slipped on a smooth round cone with half-angle θ. Find the tension of the chain if it rotates with a constant angular velocity ω about a vertical axis coinciding with the symmetry axis of the cone.

A) $T = \frac{5mg}{3\pi}\left(\cot\theta + \frac{2\omega^2 R}{g}\right)$ B) $T = \frac{2mg}{3\pi}\left(\cot\theta + \frac{\omega^2 R}{g}\right)$

C) $T = \frac{mg}{2\pi}\left(\cot\theta + \frac{\omega^2 R}{g}\right)$ D) $T = \frac{mg}{\pi}\left(\cot\theta + \frac{\omega^2 R}{2g}\right)$

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4. The time period of a body undergoing simple harmonic motion is given by $T = p^a D^b S^c$, where p is pressure, D is density, and S is surface tension. The values of a, b and c respectively are

A) $1, \frac{1}{2}, \frac{3}{2}$ B) $\frac{3}{2}, -\frac{1}{2}, 1$ C) $1, -\frac{1}{2}, \frac{3}{2}$ D) $-\frac{3}{2}, \frac{1}{2}, 1$

5. A standing sound wave in a pipe has five displacement nodes and five antinodes. The harmonic number for the standing wave is

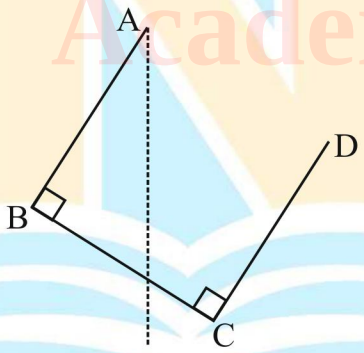
A) 3 B) 4 C) 5 (D) 9

6. Two ends of a rod of uniform cross-sectional area are kept at temperature $3T_0$ and T_0 as shown in figure. Thermal conductivity of rod varies as $K = \alpha T$, where α is a constant and T is absolute temperature. In steady state, the temperature of the middle section of the rod is (curved surface of rod is thermally insulated)



A) $\sqrt{7}\,T_0$ B) $\sqrt{3}\,T_0$ C) $2\,T_0$ D) $\sqrt{5}\,T_0$

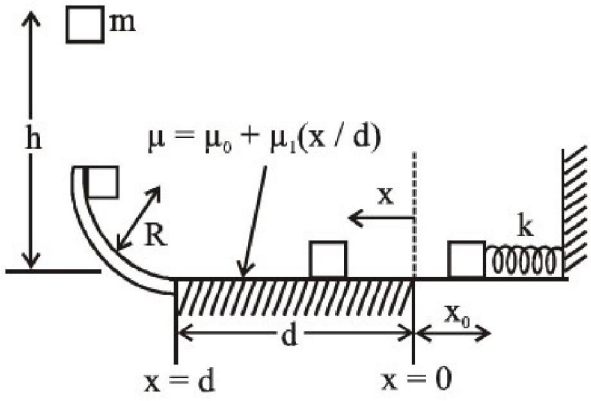
7. Three identical uniform rods are joined and hinged at A as shown in figure. The angle (in radian) made by the rod AB with the vertical in equilibrium is



A) $\sin^{-1}\left(\frac{4}{5}\right)$ B) $\tan^{-1}\left(\frac{4}{3}\right)$ C) $\tan^{-1}\left(\frac{3}{4}\right)$ D) $\tan^{-1}\left(\frac{2}{3}\right)$

8. An object of mass m is released from an initial state of rest from a spring of constant k that has been compressed a distance x_0. After leaving the spring (at the position x = 0 when the spring is relaxed) the object travels a distance d along a horizontal track that has a coefficient of friction that varies with position as $\mu = \mu_0 + \mu_1 (x/d)$. Following the horizontal track, the object enters a quarter turn of a frictionless loop whose radius is R. Finally, after exiting the quarter turn of the loop the object travels vertically upward to a maximum height, h (as measured from the horizontal surface). Maximum height (h) attained by the object is

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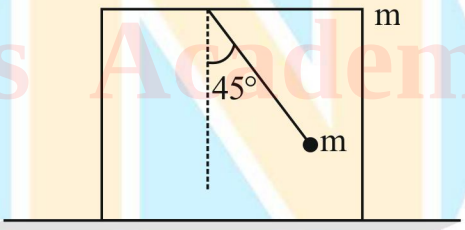
A) $\frac{kx_0^2}{2mg} - (\mu_0 + \mu_1/2)d$

B) $\frac{kx_0^2}{2mg} - (\mu_0 + \mu_1)d$

C) $\frac{kx_0^2}{mg} - (\mu_0 + \mu_1/2)d$

D) $\frac{kx_0^2}{2mg} - (2\mu_0 + \mu_1)d$

9. One end of a light string is connected to the ceiling of a box of mass m. The other end of the string is connected to a small ball of mass m and is released from the situation shown in figure. The box is placed on a smooth horizontal surface. For the situation just after release choose the correct option. (At the time of release string is in stretched condition; g is acceleration due to gravity on the surface of earth)



A) Acceleration of box is $\frac{g}{3}$

B) Acceleration of box is $\frac{g}{2}$

C) Tension in the string is $\frac{mg}{\sqrt{2}}$

D) Tension in the string is $\frac{mg}{\sqrt{3}}$

10. The time varying current is given by $i(t) = 5\sin 100\pi t + 5\cos\left(100\pi t + \frac{\pi}{6}\right)$. The rms value of the current is

A) 5A

B) $5\sqrt{2}$ A

C) $\frac{5}{\sqrt{3}}$ A

D) $\frac{5}{\sqrt{2}}$ A

11. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0(\hat{x} + \hat{y})\sin(kz - \omega t)$. It's magnetic field is given by ($\hat{x}$ and $\hat{y}$ are unit vectors along positive x and y directions respectively; c is speed of light in vacuum)

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A) $\frac{E_0}{c}(\hat{x}+\hat{y})\sin(kz-\omega t)$ B) $\frac{E_0}{c}(-\hat{x}+\hat{y})\sin(kz-\omega t)$

C) $\frac{E_0}{c}(\hat{x}-\hat{y})\sin(kz-\omega t)$ D) $\frac{E_0}{c}(\hat{x}-\hat{y})\cos(kz-\omega t)$

12. In a young's double slit experiment, one of the slits is wider than the other, so that the amplitude of the light from one slit is double of that from the other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at a phase difference ϕ is given by

A) $\frac{I_m}{3}\left(1+2\cos^2\frac{\phi}{2}\right)$ B) $\frac{I_m}{5}\left(1+4\cos^2\frac{\phi}{2}\right)$

C) $\frac{I_m}{9}\left(1+8\cos^2\frac{\phi}{2}\right)$ D) $\frac{I_m}{9}(4+5\cos\phi)$

13. A nucleus with mass number 220 initially at rest emits an α-particle. If the Q value of the reaction is 5.5 MeV, then the kinetic energy of the α-particle is

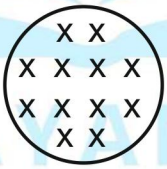
A) 4.4 MeV B) 5.4 MeV C) 5.5 MeV D) 6.5 MeV

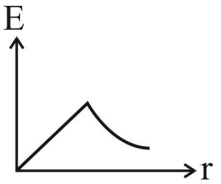
14. The flat surface of a plano-convex lens of focal length 10 cm is silvered. A point source is placed on the principal axis at a distance 30 cm in front of the curved surface and is moving with a velocity of 10 cm/s towards the lens, the velocity of image is

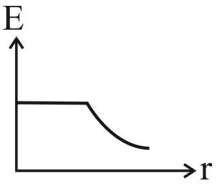
A) 10 cm/s towards the lens B) 10 cm/s away from the lens

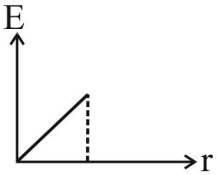
C) 0.4 cm/s away from the lens D) 0.4 cm/s towards the lens

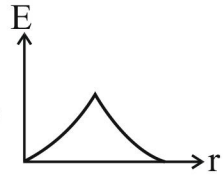
15. The magnetic field within cylindrical region whose cross-section is indicated starts increasing at a constant rate α Tesla/sec. The graph showing the variation of induced electric field with distance r from the axis of cylinder is



A) 

B) 

C) 

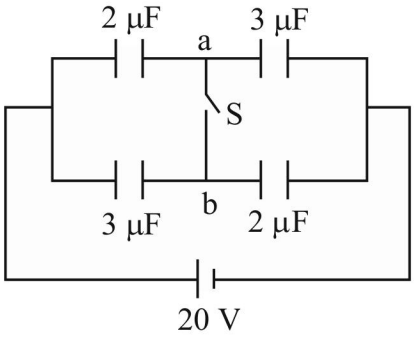
D) 

16. In a diffraction pattern of a single slit formed by light having two wavelengths λ_1 and λ_2, the first minimum with the wavelength λ_1 is coincident with the third maximum of λ_2, then

A) $3\lambda_1=\lambda_2$ B) $\lambda_1=3\lambda_2$ C) $\lambda_1=\lambda_2$ D) $2\lambda_1=7\lambda_2$

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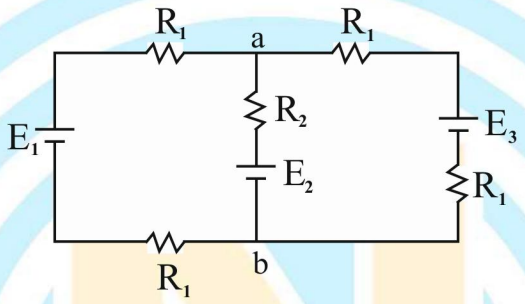
17. In figure a system of four capacitors connected across a 20 V battery is shown. Charge that will flow from switch S when it is closed is



A) 5 μC from b to a B) 5 μC from a to b

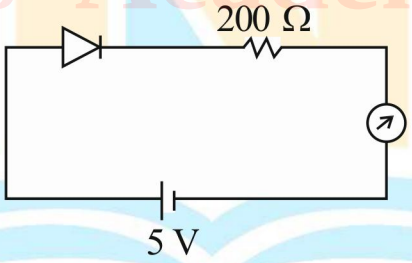
C) 10 μC from b to a D) 10 μC from a to b

18. For the circuit shown, with $R_1 = 1.0\ \Omega$, $R_2 = 2.0\ \Omega$, $E_1 = 2$ V and $E_2 = E_3 = 4$V, the potential difference between points 'a' and 'b' is approximately (in V)



A) 2.7 B) 2.3 C) 3.7 D) 3.3

19. The reading of the ammeter for a silicon diode in the given circuit is



A) 25 mA B) 11.5 mA C) 21.5 mA D) 15 mA

20. When a galvanometer is shunted with a 4Ω resistance, the deflection is reduced to one fifth. If the galvanometer is further shunted with a 2Ω wire, the further reduction in the deflection will be (the main current remains the same)

A) $\frac{5}{13}$ of the deflection when shunted with 4Ω only

B) $\frac{1}{13}$ of the deflection when shunted with 4Ω only

C) $\frac{3}{13}$ of the deflection when shunted with 4Ω only

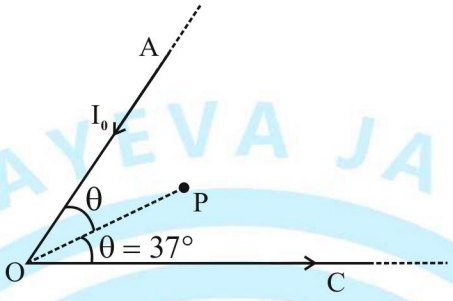
D) $\frac{3}{4}$ of the deflection when shunted with 4Ω only

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Two wires A and C carry equal current $I_0 = 1A$ as shown in figure. One end of both wires extends to infinity. The magnitude (in micro Tesla) of magnetic induction at a point P on the angle bisector of AOC at a distance 1 cm from point O is _____.



22. A beam of natural light falls on a system of 6 polaroids, which are arranged in succession such that each polaroid is turned through 30° with respect to the preceding one. The percentage of incident intensity that passes through the system will be _____.

23. The shortest wavelength of the spectral lines in the Lyman series of hydrogen spectrum is 915Å. The longest wavelength of spectral lines in the Balmer series will be _____ Å.

24. A solid sphere spinning about a horizontal axis with an angular velocity ω_0 is placed on a rough horizontal surface. Subsequently, it rolls without slipping with an angular velocity of $\frac{2\omega_0}{x}$. The value of x is _____.

25. One mole of diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is _____. (R is universal gas constant)

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. 1 mole of mixture of CO and CO_2 requires exactly 28g KOH in solution for complete conversion of all the CO_2 into K_2CO_3. How much amount more of KOH will be required for conversion into K_2CO_3. If one mole of mixture is completely oxidized to CO_2

A) 112 g B) 84 g C) 56 g D) 28 g

27. The potential energy of electron present in ground state of Li^{2+} ion is represented by

A) $\frac{+3e^2}{4\pi\varepsilon_0 r}$ B) $\frac{-3e}{4\pi\varepsilon_0 r}$ C) $\frac{-3e^2}{4\pi\varepsilon_0 r^2}$ D) $\frac{-3e^2}{4\pi\varepsilon_0 r}$

28. In which of the following arrangements, the order is NOT according to the property indicated against it?

A) Li < Na < K < Rb

Increasing metallic radius

B) I < Br < F < Cl

Increasing order of negative value of electron gain enthalpy $(-\Delta H_{eg})$

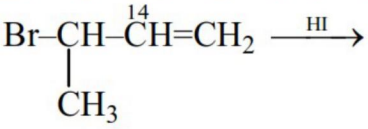
C) B < C < N < O

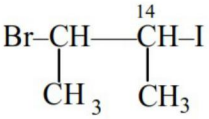
Increasing first ionization enthalpy

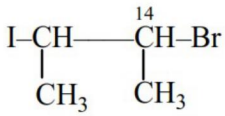
D) $Al^{3+} < Mg^{2+} < Na^+ < F^-$

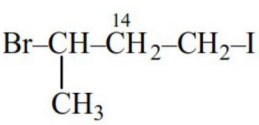
Increasing ionic size

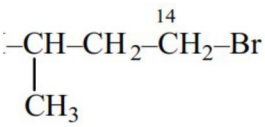
29. Products which can be obtained during the reaction in good yield



(p) 

(q) 

(r) 

(s) 

A) p, q B) q, r C) r, s D) p, r

For More Join: @IITJEE_Advanced

30. In which of the following molecules/ions BF_3, NO_2^-, NH_2^- and H_2O, the central atom is sp^2 hybridized?

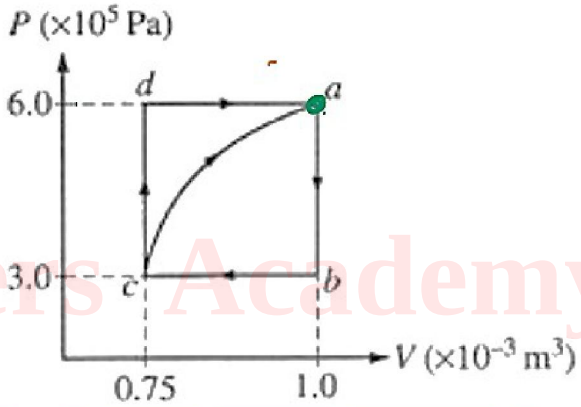
A) NH_2^- and H_2O B) NO_2^- and H_2O

C) BF_3 and NO_2^- D) NO_2^- and NH_2^-

31. The enthalpy of neutralization of a weak acid in 1 M solution with a strong base is -56.1 kJ mol^{-1}. If the enthalpy of ionisation of acid is 1.5 kJ mol^{-1} and enthalpy of neutralization of the strong acid with a strong base is -57.3 kJeq^{-1}. What is the % ionisation of the weak acid in molar solution (assume the acid is monobasic)

A) 25 B) 20 C) 15 D) 10

32. A cylinder with a movable piston contains 0.1 mole of a monoatomic ideal gas. The gas, initially at state a, can be taken through either of two cycles, abca or abcda, as shown on the PV diagram above. The following information is known about this system



$Q_{c\to a} = 685$ J along C → a path

$W_{c\to a} = -120$J along C → a path

$\Delta U_{a\to b} = 450$ J (U is internal energy)

$W_{a\to b\to c} = 75$ J

Then Heat added or removed along path b → c

A) 1090 J B) 940 J C) 1015 J D) 750 J

33. Starting with reactant only, the following equilibrium is established

$$2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$$

If $P_{Br_2} = \frac{P}{9}$, where P is total pressure at equilibrium. The ratio $\frac{K_p}{P}$ is equal to

A) $\frac{1}{3}$ B) $\frac{1}{81}$ C) $\frac{1}{9}$ D) $\frac{1}{27}$

For More Join: @IITJEE_Advanced

34. The nature of oxide (TeO_2) and hydride (TeH_2) formed by Te, respectively are

A) Reducing and basic B) Reducing and acidic

C) Oxidising and acidic D) Oxidising and basic

35. The correct statements from the following are

(a) The decreasing order of atomic radii of group 13 elements is Tl > In > Ga > Al > B.

(b) Down the group 13 electronegativity decreases from top to bottom.

(c) Al dissolves in dil. HCl and liberates H_2 but conc. HNO_3 renders Al passive by forming a protective oxide layer on the surface.

(d) All elements of group 13 exhibits highly stable +1 oxidation state.

(e) Hybridisation of Al in $[Al(H_2O)_6]^{3+}$ ion is sp^3d^2.

Choose the correct answer from the options given below

A) (a), (b), (c) and (e) only B) (c) and (e) only

C) (a), (c) and (e) only D) (a) and (c) only

36. Which is not correct statement/reaction

A) BeO is refractory material

B) $F_2 + \underset{(Ice)}{H_2O} \xrightarrow{-40^\circ C} HF + HOF$

C) $H_2SO_4 + PCl_5 \longrightarrow S_2Cl_2 + H_3PO_4$ (unbalanced)

D) $BeCO_3$ is stored in the atmosphere of $CO_2(g)$

37. Following two columns are provided. The Column-I is provided with certain complexes and column-II is provided with their features

	Column-I		**Column-II**
(i)	$[Ni(NH_3)_6]^{2+}$	(p)	d^2sp^3 hybridsation
(ii)	$[Co(H_2O)_6]^{3+}$	(q)	Outer orbital complex
(iii)	$[Cr(H_2O)_6]^{3+}$	(r)	Spin only magnetic moment is more than 2 B.M.
(iv)	$[Mn(CN)_6]^{3-}$	(s)	Effective atomic number is 36

Select the option with correct match

A) (i) → (q,r), (ii) → (q,s), (iii) → (p, r), (iv) → (p, r)

B) (i) → (q,s), (ii) → (p,s), (iii) → (p, r), (iv) → (p, r)

C) (i) → (q,r), (ii) → (p,s), (iii) → (p, s), (iv) → (p, r)

D) (i) → (q,r), (ii) → (p,s), (iii) → (p, r), (iv) → (p, r)

For More Join: @IITJEE_Advanced

38. Following two columns are provided. Column-I contains halide ions and Column-II gives their characteristics

	Column-I		Column-II
(i)	$F^{\ominus}$	(p)	Silver halide is water soluble
(ii)	$Cl^{\ominus}$	(q)	Silver halide is insoluble in aqueous ammonia solution
(iii)	$Br^{\ominus}$	(r)	Forms red gas on reaction with hot concentrated H_2SO_4
(iv)	$I^{\ominus}$	(s)	Forms red fumes on reaction with $K_2Cr_2O_7$ + concentrated H_2SO_4

Select the option with correct match

A) (i) → (p); (ii) → (s); (iii) → (r); (iv) → (q)

B) (i) → (q); (ii) → (s); (iii) → (r); (iv) → (p)

C) (i) → (p); (ii) → (r); (iii) → (s); (iv) → (q)

D) (i) → (q); (ii) → (r); (iii) → (s); (iv) → (p)

39. Select the element with correct electronic configuration

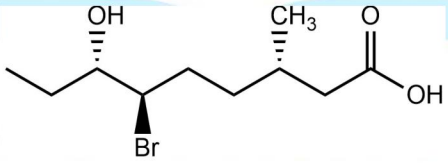
A) Th: $[Rn]5f^{1}6d^{1}7s^{2}$

B) Pu: $[Rn]5f^{7}6d^{0}7s^{1}$

C) Cm: $[Rn]5f^{7}6d^{1}7s^{2}$

D) Md: $[Rn]5f^{14}6d^{0}7s^{1}$

40. What is the correct IUPAC name for the following molecule?



A) 5-Bromo-6-hydroxy-3-methylnonanoic acid

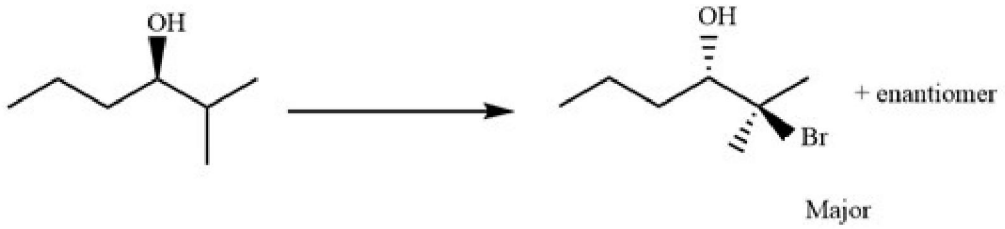
B) 4-Bromo-3-hydroxy-7-methyl-9-nonanoic acid

C) (3S ,6R ,7S)-6-Bromo-7-hydroxy-3-methylnonanoic acid

D) (3S ,6R ,7R)-6-Bromo-7-hydroxy-3-methylnonanoic acid

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41. What is the correct order of reagents needed for the following transformation?



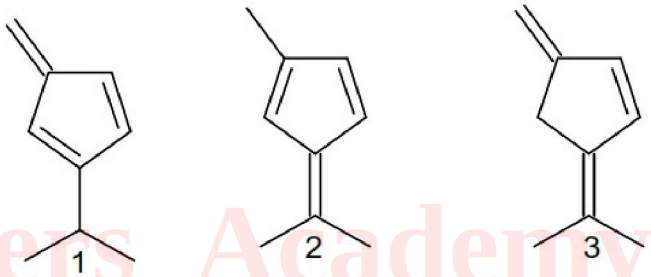
A) (1) Sodium Methoxide/Methanol, (2) MCPBA, (3) NaBr

B) (1) NaH, Ts-Cl, (2) Sodium Methoxide/Methanol, (3) MCPBA, (4) NaBr

C) (1) NaH, Ts-Cl, (2) Sodium Methoxide/Methanol, (3) MCPBA, (4) HBr

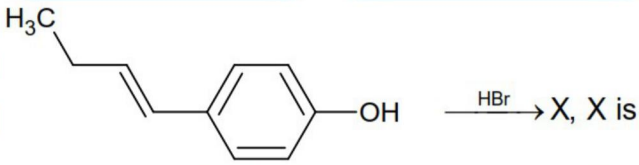
D) (1) Sodium Methoxide/Methanol, (2) MCPBA, (3) HBr

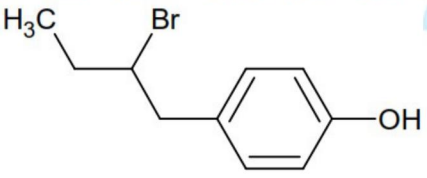
42. Which of the given compound is most acidic?

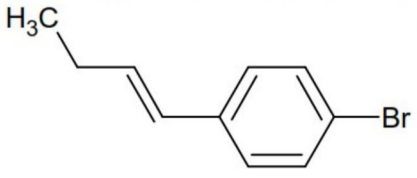


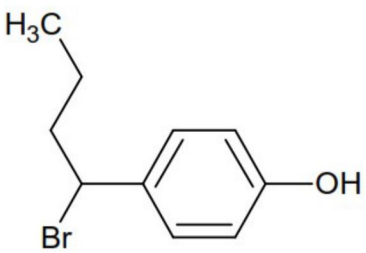
A) 1 B) 2 C) 3 D) all equal acidic

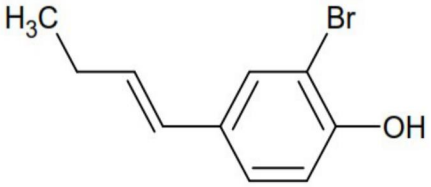
43. Identify major product



A) 

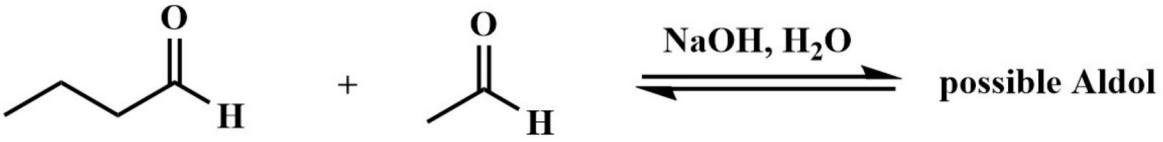
B) 

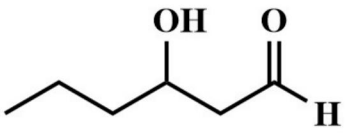
C) 

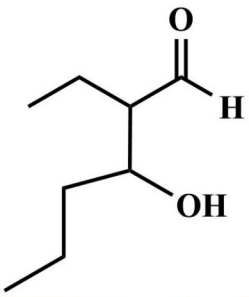
D) 

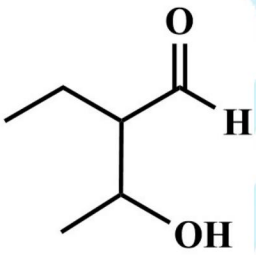
For More Join: @IITJEE_Advanced

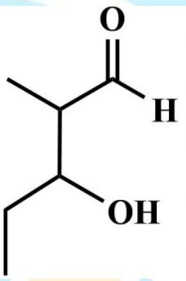
44. What is not formed as the product among the possible products of the following reaction?



A) 

B) 

C) 

D) 

45. The IUPAC name of $[Co(ONO)(NH_3)_5]Cl_2$ is

A) pentamminenitrocobalt(II) chloride

B) pentamminenitrosocobalt(III) chloride

C) pentaamminenitrito(o)cobalt(III) chloride

D) pentammineoxo-nitrocobalt (III) chloride

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The concentration of O_2 in water required to support aquatic life is 0.1×10^{-3} M. The minimum molar concentration of O_2 in the atmosphere at 27° and 1 atm pressure required to support life in water is x. Find nearest integer of 100 x.

(Given: Henry's law constant O_2 in water is 16×10^5 atm at 27°C).

$\left(R = 0.082 \text{ lt atm } K^{-1} mol^{-1}\right)$

47. Among H_2, He_2^+, Li_2, O_2, B_2, C_2, N_2, O_2^- and F_2, the number of diamagnetic species is

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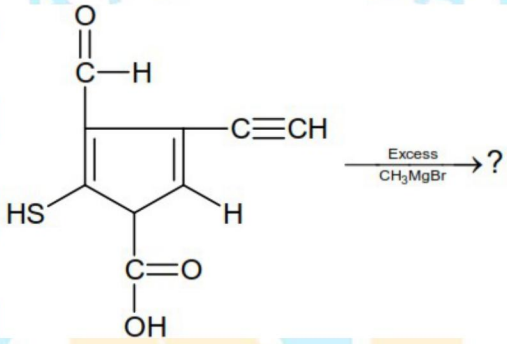
48. For the decomposition of H_2O_2 in aqueous medium the rate constant of a reaction is 0.055 min^{-1}. On decomposition of H_2O_2 the volume of O_2 collected after 20 minutes is 16 ml at STP. Find out the volume (in ml) that would be collected after complete decomposition of that sample of H_2O_2 at STP (Given ln3 = 1.1).

$\left(H_2O_2(aq) \rightarrow H_2O(l) + \frac{1}{2}O_2(g)\right)$.

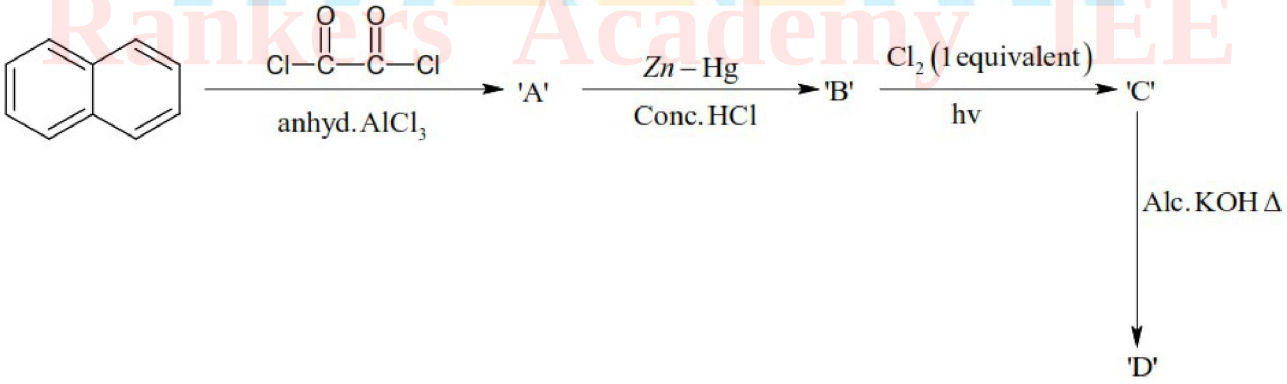
49. X = Total moles of CH_4 gas formed when one mole of the reactant reacts in the reaction

Y = Total moles of CH_3MgBr reacted when one mole of the reactant reacts in the reaction.

Calculate absolute value of (X-Y) = ?



50. The degree of unsaturation in the compound 'D' is?



(Consider major product in all reaction steps)

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let f(x) be a function which is differentiable everywhere any number of times and $f\left(2x^2-1\right)=2x^3f(x)\ \forall x\in R$ then $f^{2026}(0)$ is equal to k (where $f^n(x)$ is n^{th} derivative of f(x)), and a curve y = g(x) represented by differential equation $(2xy^2 - y)\,dx + xdy = 0$ passes through the intersection of the lines $2x - 3y = 1$ and $3x + 2y = 8$ then $|g(k+1)|$ is

A) 2 B) 0 C) 1 D) 3

52. If x = a satisfies equation $\tan^{-1}(x+2)+\cot^{-1}\sqrt{4x+20} = \lim\limits_{x\to\frac{\pi}{2}}\frac{\sin(x\cos x)}{\cos(x\sin x)}$ and

$\lim\limits_{x\to0}\frac{\sqrt[a]{1+x}-\frac{(1+x)^{1/x}}{e}-bx}{5x+kx^2+x^3}=0$ then choose the INCORRECT option

A) a = 4 B) ab = 3 C) a = 2 D) $k \in R$

53. If $f(x) = ax^2 + bx + c$ such that f(p) + f(q) = 0, where $a \neq 0$ $p, q \in R$ then number of real roots of the equation f(x) = 0 in [p, q] is

A) Exactly one B) At least one

C) At most one D) Data insufficient

54. Let $P=\lim\limits_{n\to\infty}n^{-n^2}\left\{(n+2^0)(n+2^{-1})(n+2^{-2})\ldots..(n+2^{-n+1})\right\}^n$. If nearest integer to P is denoted by k, then $\int_1^{\frac{k-1}{2}}\left[x^2-2x-2\right]dx$ is, [x] is a greatest integer function

A) –4 B) –5 C) $-\sqrt{2}-\sqrt{3}-1$ D) $-\sqrt{2}-\sqrt{3}+1$

55. Let ΔABC be isosceles triangle with AB = AC, suppose that the bisector of angle B meets the side AC at D such that BC = BD + AD, then the angle A equals (in degrees)

A) 100° B) 60° C) 40° D) 36°

56. The number of functions f : {1, 2, 3, ……. 2026} → {2025, 2026} such that f(1) + f(2) + ……+ f(2026) is an odd integer, is

A) 2^{2025} B) 2^{2026} C) $2^{2025} - 1$ D) 2^{2026}

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57. How many 4 digit number $\overline{abcd}$ are there such that $a < b < c < d$ and $b - a < c - b < d - c$?

A) 7 B) 8 C) 16 D) 9

58. If α, β, γ are the roots of the equation $x^3 - x - 1 = 0$ then $\frac{1-\alpha}{1+\alpha}+\frac{1-\beta}{1+\beta}+\frac{1-\gamma}{1+\gamma}$ is

A) 1 B) -1 C) 0 D) 2

59. Let $\vec{v}_1$ and $\vec{v}_2$ are two vectors such that $\vec{v}_1 = 2(\sin\alpha+\cos\alpha)\hat{i}+\hat{j}$ and $\vec{v}_2 = \sin\beta\,\hat{i}+\cos\beta\,\hat{j}$ where α and β satisfy the relation $2(\sin\alpha+\cos\alpha)\sin\beta = 3-\cos\beta$, let the value of $(3\tan^2\alpha + 4\tan^2\beta)$ is given by k_1 and the variance of 10 natural numbers 1, 1, 1, ……, 1, k is less than 10, if maximum possible value of k is k_2 then choose the correct option

A) $3k_2 = k_1 - 2$ B) $3k_1 = k_2 - 2$ C) $2k_2 = k_1 - 3$ D) $2k_1 = k_2 - 3$

60. Two concentric ellipses are such that the foci of one lie on the another and the length of their major axes are equal, if e_1 and e_2 be their eccentricities. If angle between their axes is θ, then

A) $\cos\theta = \sqrt{\frac{1}{e_1^2}+\frac{1}{e_2^2}-\frac{1}{e_1^2e_2^2}}$ B) $\sin\theta = \sqrt{\frac{1}{e_1^2}+\frac{1}{e_2^2}-\frac{1}{e_1^2e_2^2}}$

C) $\cos\theta = \sqrt{\frac{1}{e_1^2}+\frac{1}{e_2^2}-\frac{2}{e_1^2e_2^2}}$ D) $\sin\theta = \sqrt{\frac{1}{e_1^2}+\frac{1}{e_2^2}-\frac{2}{e_1^2e_2^2}}$

61. The sum of $\frac{3}{1!+2!+3!}+\frac{4}{2!+3!+4!}+\ldots\ldots+\frac{2026}{2024!+2025!+2026!}$, is equal to

A) $\frac{1}{2}-\frac{1}{2026!}$ B) $\frac{1}{2}-\frac{1}{2024!}$

C) $\frac{1}{2024!}-\frac{1}{2026!}$ D) $\frac{1}{2025!}-\frac{1}{2026!}$

62. The fraction exceeding its own n^{th} power ($n \in N$) by maximum possible value is

A) $\left(\frac{1}{n}\right)^{\frac{1}{n-1}}$ B) $\left(\frac{1}{n}\right)^{n-1}$ C) $\left(\frac{1}{n}\right)^{n}$ D) $\left(\frac{n}{n+1}\right)^{\frac{1}{n-1}}$

63. Largest possible real part of $(75+117i)Z+\frac{96+144i}{Z}$ and $|Z| = 4$, (Z is a complex number) is k then sum of the digits of k is

A) 8 B) 9 C) 11 D) 13

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64. Let U be the set {1, 2, 3, ……. n} and $A_1, A_2, \ldots A_m$ be k element subsets of U. It is given that for every pair of elements in U there are exactly x subsets which contains both the elements, then in how many subsets the element 1 is contained?

A) $\frac{k(k-1)m}{n(n-1)}$ B) $\frac{km}{n}$ C) $\frac{(k-1)m}{(n-1)}$ D) $\frac{km}{n-1}$

65. Suppose that a parabola has vertex $\left(\frac{1}{4}, -\frac{9}{8}\right)$ and equation $y = ax^2 + bx + c$, $a > 0$ and $a + b + c$ is an integer. The minimum possible positive value of a can be written in the form of $\frac{p}{q}$, where p and q are integers and relatively prime then p + q is

A) 8 B) 12 C) 5 D) 11

66. In ΔABC where B = (0, 0), C = (56, 0) and A = (28, α) α > 0 such that AC = 100. A circle P has radius 16 and is tangent to AC and BC, circle Q is externally tangent to P and is tangent to AB and BC. No point of circle Q lies outside the triangle ABC. The radius of circle Q can be expressed in the form of $m - n\sqrt{k}$ where m, n and k are positive integers and k is the product of distinct primes, then m + nk is equal to

A) 204 B) 231 C) 254 D) 23

67. It is given that f(x) is function defined on R, satisfying f(1) = 1 and for any $x \in R$ $f(x + 5) \geq f(x) + 5$ and $f(x + 1) \leq f(x) + 1$, if g(x) = f(x) + 1 − x then g(2026) is

A) 2025 B) 2026 C) 0 D) 1

68. Let $M_n = \{0.a_1a_2\ldots\ldots a_n \mid a_i = 0 \text{ or } 1, 1 \leq i \leq n-1 \text{ and } a_n = 1\}$ be a set of decimal fractions. T_n and S_n be the number and the sum of the elements of M_n respectively. If $\lim_{n\to\infty} \frac{S_n}{T_n}$ can be written as $\frac{a}{b}$ (a and b are relatively prime) then a + b is equal to

A) 19 B) 18 C) 17 D) 16

69. The rule of an "obstacle course" specifies that at n^{th} obstacle a person has to toss a die n times, if the sum of points in these n tosses is bigger than 2^n, then person is said to have crossed the obstacle, then the probability that Jaspreet Bumrah crosses the first three obstacles, is

A) $\frac{20}{27}$ B) $\frac{100}{243}$ C) $\frac{143}{243}$ D) $\frac{151}{243}$

For More Join: @IITJEE_Advanced

70. Let $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ are four position vectors in space such that $|\overrightarrow{AB}| = 3$ $|\overrightarrow{BC}| = 7$ $|\overrightarrow{CD}| = 11$ and $|\overrightarrow{DA}| = 9$ then the value of $|\overrightarrow{AC}| \cdot |\overrightarrow{BD}|$ has

A) Only one value
B) two values
C) four values
D) infinite many values

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $a_0, a_1, a_2, \ldots\ldots, a_n$ be a sequence of numbers satisfying $(3 - a_{n+1}) \cdot (6 + a_n) = 18$ and $a_0 = 3$ then $\sum_{i=0}^{2026} \frac{1}{a_i}$ is given by $\frac{1}{k_1}\left(2^{k_2} - (k_2 + 1)\right)$ then $k_1 . k_2$ is _____.

72. Suppose $x, y \in (-2, 2)$ and $xy = -1$ then minimum value of $\frac{4}{4 - x^2} + \frac{9}{9 - y^2}$ is _____.

73. Two functions $y = f(x)$ and $y = g(x)$ are continuous whose graphs intersect at three points $(0, 4), (2, 2)$ and $(4, 0)$ with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 (f(x) - g(x))dx = 10$ and $\int_2^4 (g(x) - f(x))dx = 5$ then the area between two curves in $0 < x < 2$, is _____.

74. If 2d be the shortest distance between the lines $\frac{y}{b} + \frac{z}{c} = 1; x = 0$ and $\frac{x}{a} - \frac{z}{c} = 1; y = 0$ $(a, b, c > 0)$ such that $d = \frac{k.abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}$ then positive value of the determinant of the matrix A whose $\text{Adj}(\text{Adj } A) = \begin{bmatrix} 13+k & k+27 & -15+k \\ -15+k & 13+k & k+27 \\ k+27 & -15+k & 13+k \end{bmatrix}$ is

75. If $I = \int_0^1 \frac{2 + 3x + 4x^2}{2\left(\sqrt{1 + x + x^2}\right)} dx$, then $I^2 + 19$ is _____

For More Join: @IITJEE_Advanced